

General Concepts

General Terms

- Bandwidth**: max data rate of a link (bits/sec).
- Goodput**: proportion of successful slots (transmission without collision).
- Throughput**: proportion of slots with a transmission (success or not).
- Propagation time**: time for a signal to travel between two nodes.

Mathematical Foundations

Probability

- Conditional**:  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ ,  $P(B) > 0$
- Bayes**:  $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Total Prob. Law**:  $P(B) = \sum_i P(B|A_i)P(A_i)$  (partition  $\{A_i\}$ )
- Expected Value**:  $\mathbb{E}[X] = \sum_x xP(X=x)$
- Memoryless**:  $P(X > s+t | X > t) = P(X > s)$  (Geo/Exp).

| Dist.     | PMF                                 | Notation               | $\mathbb{E}[X]$ | $\text{Var}(X)$   |
|-----------|-------------------------------------|------------------------|-----------------|-------------------|
| Bernoulli | $P(X=1)=p$                          | $\text{Ber}(p)$        | $p$             | $p(1-p)$          |
| Binomial  | $\binom{n}{k} p^k (1-p)^{n-k}$      | $\text{Bin}(n, p)$     | $np$            | $np(1-p)$         |
| Geometric | $(1-p)^{k-1}p$                      | $\text{Geo}(p)$        | $\frac{1}{p}$   | $\frac{1-p}{p^2}$ |
| Poisson   | $\frac{\lambda^k e^{-\lambda}}{k!}$ | $\text{Pois}(\lambda)$ | $\lambda$       | $\lambda$         |

Computer networks vs Distributed systems

Circuit and Packet Switching

Timing

Multiplexing

The OSI Model (5-Layer model)

- Physical**: Move bits on a link (signals, timing, encoding).
- Link**: Frames on local network, MAC addressing, error detection.
- Network**: Packet delivery across networks, routing (IP).
- Transport**: End-to-end process communication, reliability/flow control (TCP/UDP).
- Application**: End-user services/protocols (DNS, DHCP, HTTP, FTP etc.).

Link Layer & Local Area Networks (LAN)

Shared Medium Intuition

- Nodes broadcast on same channel; only one can transmit at a time.
- Simultaneous transmit  $\Rightarrow$  collision  $\Rightarrow$  data lost.
- Sender uses full link bandwidth while transmitting.

Definitions

- Bandwidth**: max data rate of channel ( $B$ ).
- Throughput**: fraction of bandwidth used to send traffic.
- Goodput**: fraction of bandwidth used for successful data.
- $\eta = \frac{\text{avg effective bandwidth}}{\text{total bandwidth}} = \frac{\text{avg effective time}}{\text{total time}}$ ,  $0 \leq \eta \leq 1$ .

Multiple Access Domains

- Multiple Access Domain**: set of nodes sharing a common channel.
- Channel Partitioning**: divide channel into smaller "pieces" (time/freq/code).
- Random Access**: transmit at will; handle collisions (ALOHA, CSMA).
- Taking Turns**: nodes take turns to transmit (polling/token passing).

ALOHA protocol

- Random access on shared medium; collisions if overlap. Frame time  $T$  (size  $L$ , rate  $B$ ):  $T = L/B$ .
- Binomial model**:  $N$  users, each transmits in a slot w.p.  $p$ .
- Poisson model**: aggregate attempts are Poisson with rate  $G$  (attempts/slot).
- Slotted ALOHA (align to slots)**:
  - Success if exactly one tx in a slot.
- Binomial:  $P(\text{succ}) = Np(1-p)^{N-1}$ , max at  $p = 1/N$  gives  $S_{\max} \approx 1/e$ .
- Poisson:  $S = Ge^{-G}$ ,  $S_{\max} = 1/e$  at  $G = 1$ .
- Pure ALOHA (no slots)**:
  - Vulnerable period  $2T$ .
- Binomial (approx):  $P(\text{succ}) \approx Np(1-p)^{2(N-1)}$ , max at  $p \approx 1/(2N)$  gives  $S_{\max} \approx 1/(2e)$ .
- Poisson:  $S = Ge^{-2G}$ ,  $S_{\max} = 1/(2e)$  at  $G = 1/2$ .

CSMA (Carrier Sense Multiple Access)

- Listen before tx; if channel idle, transmit entire frame; else wait.
- Propagation delay  $\tau$** : time for signal to propagate between 2 farthest nodes.
- CSMA/CD (Collision Detection)**: abort tx on collision detection; reduces wasted time.
- CSMA/CD efficiency**:  $\eta = \frac{1}{1+5 \frac{d_{\text{prop}}}{d_{\text{trans}}}}$ .

CSMA/CD (Collision Detection)

- Detect collisions while transmitting; send jam; then backoff and retry.
- Binary exponential backoff**: after  $m$ -th collision pick  $k \in \{0, \dots, 2^m - 1\}$ ; wait  $k$  slots.

CSMA/CA (Collision Avoidance)

- Used in Wi-Fi: cannot detect collisions reliably, so avoid them.
- If channel idle, wait DIFS then transmit; if busy, pick random backoff slots.
- Receiver replies with ACK after SIFS; missing ACK implies collision.

CSMA persistence

- 1-persistent**: transmit immediately when channel becomes idle (high collision under load).
- 0-persistent**: wait random backoff after busy (lower collision, higher delay).
- p-persistent**: when idle, transmit with prob.  $p$  each slot (tradeoff).

Single (Logical) Link in practice

MAC Addresses

- Link-layer address unique per NIC (Network Interface Card); assigned by IEEE OUI.
- Used to deliver frames on same LAN; flat (non-hierarchical), portable across LANs.
- Usually burned-in, but can be software-set.

Ethernet

- Dominant wired LAN; many speed generations; cheap NICs.
- CSMA/CD (classic shared medium)**:
  - Sense channel; if idle transmit, else wait.
  - If collision detected while tx, abort and send jam.
  - Binary exponential backoff**: after  $m$ -th collision pick  $k \in \{0, \dots, 2^m - 1\}$ ; wait  $k \cdot 512$  bit-times, retry.

Interconnecting broadcast domains

- Switch**:
  - link-layer device that breaks broadcast domains; stores/forwards frames using MAC table.
  - Transparent to hosts; self-learning (no manual config), entries age out (TTL).
  - Each link is its own collision domain; enables simultaneous transmissions.
  - Forwarding: if dest known on incoming port  $\Rightarrow$  drop; else forward to port; if unknown  $\Rightarrow$  flood.
- Switch table**: (MAC, port, TTL). Learn from source MAC+ingress port on each frame.
- Loops**: cause broadcast storms, multiple frame copies, MAC table instability.
- Spanning Tree Protocol (STP)**: disable links to eliminate loops; elect root, compute shortest-path tree.
- HELLO/BPDU frames**: SID (switch ID), RID (root ID), DTR (distance to root).
- Root selection: prefer lower RID; then lower DTR; then lower SID (tie-break).

"Layer 2.5" - ARP

ARP is generally considered to be part of Layer 2 (Data Link Layer), but it works directly to support Layer 3 (Network Layer).

- Why it's Layer 2: ARP messages are encapsulated directly into Ethernet frames. They do not have an IP header (Layer 3) and they cannot be routed outside of your local network (LAN).
- Why it feels like Layer 3: Its entire purpose is to handle Layer 3 information (IP addresses). It is the mechanism that allows Layer 3 to actually function on physical hardware.

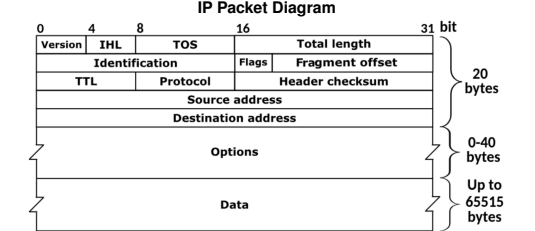
Address Resolution Protocol (ARP)

- Translates between MAC addresses and IP addresses on a LAN.
- Needed because Layer 3 provides the IP, but frames on Layer 2 need a destination MAC.
- To send outside the LAN, the packet uses the IP of the destination but the MAC of the next-hop router; routers then forward using their own L2 on each hop.
- Hosts use DHCP and routing to know whether the destination IP is on the same LAN or remote.
- Each node maintains an **ARP cache** (IP  $\rightarrow$  MAC table).
- If the mapping is missing, send an **ARP request** (broadcast in the local broadcast domain).
- The target replies with an **ARP response** containing its MAC, which is stored in the cache.
- ARP messages stay within a single broadcast domain (not routed).

Network Layer (IP & Routing)

Introduction to IP addressing

- IPv4 address = 32 bits (a.b.c.d), identifies interface (host/router can have multiple interfaces).
- Subnet vs host split via CIDR: a.b.c.d/x where  $x$  = prefix length (subnet mask).
- Example: 192.168.1.178/24  $\Rightarrow$  subnet 192.168.1, host 178.
- Hierarchical addressing enables routing scalability; ISPs advertise prefixes.
- Forwarding uses **longest prefix match**.
- Address discovery protocols**:



Transport Layer (TCP & UDP)

TCP (Transmission Control Protocol)

- Reliable, in-order byte stream; receiver ACKs received data. Congestion control limits the sending rate.
- vs. UDP: UDP adds port+IP multiplexing with no reliability guarantees.
- In each ACK, the acknowledgment number is the next expected byte (largest in-order byte + 1).
- MTU** (max transfer unit), **MSS** (max segment size); typically  $MSS < MTU$ .

Connection setup (three-way handshake)

- Client sends **SYN** with initial sequence number  $M$ .
- Server replies **SYN+ACK** with seq  $N$  and ACK  $M+1$ .
- Client sends **ACK** with ACK  $N+1$ .

Connection teardown

- Client sends **FIN**.
- Server replies **ACK**.
- Server sends **FIN**.
- Client replies **ACK**, waits **timeout**, then closes.

Reliable data transfer (sender)

- Maintain sequence number and buffer; segment and send data.
- Send while window allows:  $lastByteSent - lastByteAcked \leq RW$  (receiver window).
- Start timer if not running; on timeout, retransmit the oldest unACKed segment.

Reliable data transfer (receiver)

- Deliver in-order data and send cumulative ACK.
- If out-of-order segment arrives, send ACK for last in-order byte (duplicate ACK).

Retransmissions

- Retransmit on either **timeout** or **duplicate ACKs**.
- Fast retransmit**: many duplicate ACKs imply loss before timeout.

RTT estimation

- SampleRTT*: time from sending a segment until its first ACK returns.
- EstimatedRTT<sub>i</sub>* =  $(1 - \alpha)EstimatedRTT_{i-1} + \alpha SampleRTT$ , with  $\alpha = 0.125$ .
- DevRTT<sub>i</sub>* =  $(1 - \beta)DevRTT_{i-1} + \beta |SampleRTT - EstimatedRTT_{i-1}|$ .
- TimeoutInterval* = *EstimatedRTT* + 4 *DevRTT*.

Flow Control vs. Congestion Control

- Flow control**: prevents sender from overwhelming the receiver (receiver buffer limits).
- Congestion control**: prevents sender from overloading the network.
- Sender can send while:  $lastByteSent - lastByteAcked \leq \min(RW, cwnd)$ .
- When receiver buffer is the limiter, flow control dominates; when network is the limiter, congestion control dominates.

Congestion Control phases

- Slow Start (SS)**: exponential growth; for each ACK,  $cwnd = cwnd + 1$  (per ACK), doubling each RTT.
- Congestion Avoidance (CA)**: linear growth; for each ACK,  $cwnd = cwnd + 1/cwnd$  (+1 per RTT).
- Threshold *sssthresh* decides SS vs CA: use SS while  $cwnd < ssthresh$ , CA when  $cwnd \geq ssthresh$ .

TCP Tahoe

- Start with  $cwnd = 1$  (SS).
- On loss (timeout or 3 duplicate ACKs): set  $ssthresh = cwnd/2$ , set  $cwnd = 1$ , restart SS.

TCP Reno

- Improves Tahoe for 3 duplicate ACKs (fast retransmit/fast recovery).
- On 3 duplicate ACKs: set  $ssthresh = cwnd/2$  and enter **fast recovery** (do not go back to  $cwnd = 1$ ).
- During fast recovery: for each duplicate ACK, increase  $cwnd$  by 1; on new ACK, set  $cwnd = ssthresh$  and enter CA.
- On timeout: set  $ssthresh = cwnd/2$ , set  $cwnd = 1$ , restart SS.

TCP Vegas

- Delay-based congestion control: compares expected throughput (from  $cwnd/baseRTT$ ) to actual throughput (from  $cwnd/RTT$ ).
- Adjusts  $cwnd$  to keep the difference within a target range ( $\alpha, \beta$ ), aiming to prevent queue buildup before loss.
- Less widely used because it competes poorly with loss-based flows (Reno/CUBIC), is sensitive to RTT measurement noise, and offers limited incremental deployability in the public Internet.

Utilization intuition

- With window size  $W$  and RTT  $RTT$ , throughput is about  $W/RTT$ .
- If window halves after loss, average throughput over a cycle is roughly  $0.75 W/RTT$ .

UDP (User Datagram Protocol)

- Connectionless; no setup, no reliability, no ordering.
- Uses IP+port for multiplexing/demultiplexing (socket identified by destination port).
- Very small header; no congestion control, so can send as fast as the application wants.
- Useful when timeliness matters more than reliability (e.g., streaming, VoIP), or when simplicity/overhead is key.
- Also used for request/response protocols like DNS and for DHCP (no TCP connection possible before address assignment).
- Reliability (if needed) is handled at the application layer.

Application Layer (HTTP, DNS, SMTP & FTP)

DNS (Domain Name System)

Maps between hostnames and IP addresses. Each server is authoritative for its domain and can resolve queries or forward them; resolution ends when an authoritative server replies or the requested data is found in cache.

Record types:

- A**: maps an Internet hostname to an IP address.
- NS**: maps a domain name (e.g., google.com) to the server responsible for it.
- CNAME**: maps a hostname to its canonical name, e.g., wedsac.mit.com is actually www.mit.com.
- PTR**: reverse of A (IP address to hostname).
- MX**: maps a hostname to the mail server that handles it.

DNS queries and replies use UDP and typically port 53 (smaller packets for requests/replies). Servers return a **TTL**; the answer is cached for that duration.

**Glued response**: if the client needs to query the authoritative server but does not know its IP, the server returns the IP in the NS response.

Security issues:

- DDoS Attack**: flooding DNS servers so they cannot answer other requests.
- MITM (Man-in-the-Middle) Attack**: when a client queries a server, a fake response can arrive before the real one, potentially mapping a requested name to a desired IP.
- Cache poisoning**: if the server is not well-protected, it continues serving an injected response until TTL expires; thus, we must ensure the reply is authentic.
- Kaminsky attack**: attacker sends many requests for random subdomains of a target domain (e.g., random1.target.com, random2.target.com, etc.) to a resolver; when the resolver queries the authoritative server, the attacker floods it with fake replies, hoping one matches the query ID and port, poisoning the cache.
- DNSSEC**: adds signatures so you can verify who answered and add more security; adds extra information and expands UDP packet size.

DHCP (Dynamic Host Configuration Protocol)

- Allows host to obtain IP dynamically (no manual config).
- Each time host connects, it can get a different IP; server can **lease** addresses and reuse later.
- Used by devices joining a network.
- Message flow (DORA)**:
  - DHCP Discover**: src IP 0.0.0.0, dst 255.255.255.255 (broadcast, server unknown).
  - DHCP Offer**: server replies, dst 255.255.255.255 (broadcast); includes **yiaddr** (your IP).
  - DHCP Request**: client broadcasts 0.0.0.0  $\rightarrow$  255.255.255.255 to accept one offer; includes chosen **yiaddr** so other servers withdraw.
  - DHCP ACK**: server confirms lease; may broadcast so others know address in use.
- DHCP relaying**: if DHCP server not on local subnet, relay forwards messages to server on another network.

Interdomain Routing & BGP

Autonomous Systems (AS)

- Definition**: IP networks/routers under one admin with a common routing policy.
- ASN**: Unique 16/32-bit AS identifier used in BGP.
- AS relationships**:
  - Customer / Provider**: Customer pays provider for transit.
  - Full / Partial transit**: Full = global reachability; partial = limited reach.
  - Peering**: Settlement-free; advertise own + customers only.
- AS-level topology**: Common categories
  - Tier-1**: No providers; global reach via peering.
  - Tier-2**: Providers + peers (buys some transit).
  - Stub AS**: Single provider; no downstream customers.

Border Gateway Protocol (BGP)

- Definition**: Inter-AS routing protocol.
- Path-vector**: Advertises AS\_PATH; reject if own ASN appears.
- Policy + attributes**: LOCAL\_PREF, MED, AS\_PATH, NEXT\_HOP, etc.

BGP message flow (5 stages)

- TCP setup**: Session on TCP/179.
- OPEN**: Exchange ASN, hold-time, capabilities.
- Initial UPDATE**: Send NLRI + attributes.
- Decision**: Import policy + best-path selection.
- Steady state**: KEEPALIVE + incremental UPDATE/NOTIFICATION.

iBGP, eBGP and IGP interaction

- eBGP**: Between ASes; increments AS\_PATH, may change NEXT\_HOP.
- iBGP**: Within AS; no AS\_PATH prepend; needs full mesh or route-reflectors.
- IGP**: Provides internal reachability to BGP NEXT\_HOP.

BGP Stability

- Stable state**: Routes converge with no further changes.
- Safety theorem (Informal)**: Under Gao-Rexford, BGP converges to a stable outcome.
- Gao-Rexford conditions (commercial routing)**:
  - Topology**: Customer-provider edges form a DAG; peers are lateral.
  - Preference**: Customer > peer > provider.
  - Export**: Export all routes to customers; only customer routes to peers/providers.